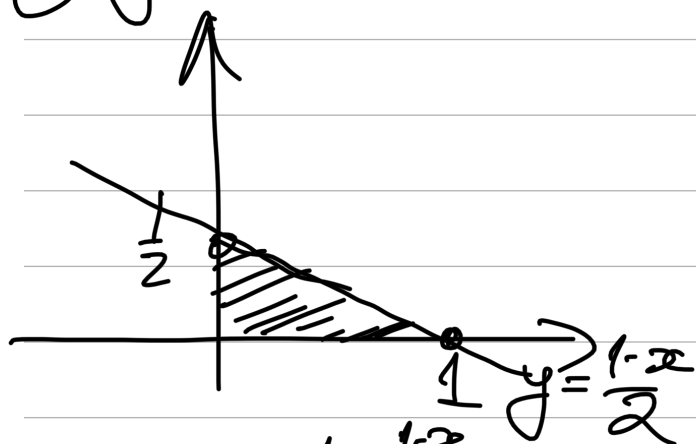


300.1



$$\begin{cases} x, y \geq 0 \\ y \leq \frac{1-x}{2} \Leftrightarrow x \leq 1-2y \end{cases}$$

$$a) 1 = \int_0^1 \int_0^{\frac{1-x}{2}} (cx^2 + 1) dy dx = \frac{1}{2} \int_0^1 (cx^2 + 1)(1-x) dx$$

$$= \frac{1}{2} \int_0^1 (cx^2 + 1 - cx^3 - x) dx$$

$$= \frac{1}{2} \left[c \frac{1}{3} + 1 - c \frac{1}{4} - \frac{1}{2} \right]$$

$$= \frac{c}{24} + \frac{1}{4} \Rightarrow c = \frac{3}{4} \cdot 24 = 18.$$

$$f_X(x) = \int_0^{\frac{1-x}{2}} (cx^2 + 1) dy = [cx^2 + 1] \cdot \left(\frac{1-x}{2}\right)$$

$$EY = \int_0^1 \int_0^{\frac{1-x}{2}} y (cx^2 + 1) dy dx$$

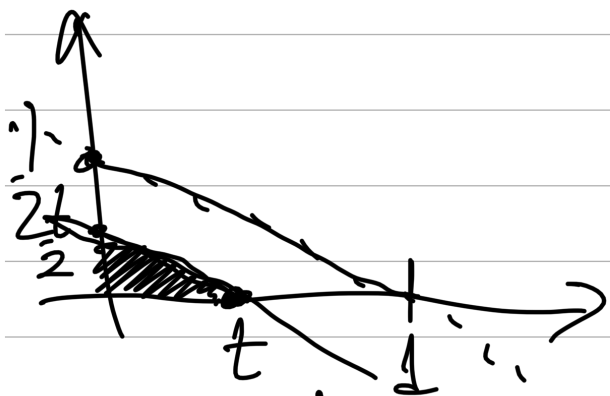
$$\begin{aligned}
&= \int_0^1 (cx^2 + 1) \cdot \left(\frac{1-x}{2}\right)^2 \cdot \frac{1}{2} dx \\
&= \frac{1}{8} \int_0^1 (cx^2 + 1)(1 - 2x + x^2) dx \\
&= \frac{1}{8} \int_0^1 cx^2 + 1 - 2cx^3 - 2x + cx^4 + x^2 dx \\
&= \frac{1}{8} \left[6 + 1 - 9 - 1 + \frac{18}{5} + \frac{1}{3} \right] \\
&= \frac{1}{8} \left(\frac{54 + 5 - 45}{15} \right) = \frac{14}{120} = \frac{7}{60}
\end{aligned}$$

$$d) f_{Y|X}\left(y \middle| \frac{1}{2}\right) = \frac{f_{X,Y}\left(\frac{1}{2}, y\right)}{f_X\left(\frac{1}{2}\right)} = \frac{c\left(\frac{1}{2}\right)^2 + 1}{\left(\left(\frac{1}{2}\right)^2 + 1\right)\left(\frac{1-\frac{1}{2}}{2}\right)} = 4$$

$$\begin{aligned}
&\exists a \quad y \in (0, \frac{1}{2}) \Rightarrow \mathbb{P}[Y|X = \frac{1}{2}] = \mathbb{P}[U(0, \frac{1}{2})] \\
&= \frac{1}{8}
\end{aligned}$$

$$B) \exists a \quad z \in (0, 1)$$

$$P(X+2Y \leq t) = P(Y \leq \frac{t-X}{2})$$



$$= \int_0^{t/2} \int_0^{t-2y} c x^2 + 1 dx dy$$

$$= \int_0^{t/2} c \left[\frac{(t-2y)^3}{3} + t-2y \right] dy$$

$$\stackrel{w=t-2y}{=} \frac{1}{2} \int_0^t c \left[\frac{w^3}{3} + w \right] dw$$

$$= \frac{1}{2} \left[18 \cdot \frac{t^4}{12} + \frac{t^2}{2} \right]$$

$$= \frac{1}{4} (3t^4 + t^2)$$

$$\Rightarrow f_2(t) = 3t^3 + \frac{1}{2}t \quad \text{for } t \in (0,1).$$

u. Dursse.

Um cbc nutzen: $\begin{vmatrix} Z = X+2Y \\ W = X \end{vmatrix} \Rightarrow \begin{vmatrix} X = W \\ Y = \frac{Z-W}{2} \end{vmatrix}$

$$|J| = \begin{vmatrix} 1 & 0 \\ -\frac{1}{2} & 1 \end{vmatrix} = \frac{1}{2}$$

$$\Rightarrow f_{Z,W}(z,w) = f_{X,Y}\left(w, \frac{z-w}{2}\right) \cdot \frac{1}{2} = (w^2 + 1) \cdot \frac{1}{2}$$

$$\exists^\alpha w, \frac{z-w}{2} \in (0,1) \text{ u } z + 2\frac{z-w}{2} \leq 1,$$

$$\text{i.e. } z \in (0,1) \text{ u } w \in (0,z)$$

$$\Rightarrow f_Z(t) = \int_0^t (w^2 + 1) \cdot \frac{1}{2} dw = \frac{1}{3}t^3 + \frac{t}{2}$$

$$\exists^\alpha t \in (0,1).$$

200.2 Herea broenata niy kullerata uog 30 ca
 $X_1, X_2, \dots \sim \text{Exp}(\frac{1}{5})$ u a p. d. r.
 Kullerata uog 30 - $Y_1, Y_2, \dots \sim \text{Exp}(\frac{1}{10})$
 $EX_1 = 5, DX_1 = 25$

$$a) P(X_1 + \dots + X_{100} < 480)$$

$$= P\left(\frac{X_1 + \dots + X_{100} - 100 \cdot 5}{\sqrt{25} \cdot \sqrt{100}} < -\frac{20}{\sqrt{25} \cdot 10}\right)$$

$$\approx \Phi(-0,4) \approx 34,458\%$$

$$5) Z_{i=1:n} (X_1, Y_1) \sim \text{Exp}\left(\frac{1}{5} + \frac{1}{10}\right) = \text{Exp}\left(\frac{3}{10}\right)$$

Б) Аво Z_i е ~~бавно~~ $\text{arg}(i-1)$ -ва n $i=1, \dots, n$
~~2000~~, $Z_i \sim \text{Exp}\left(\frac{3}{10}\right)$

$$EZ_i = \frac{10}{3}, \quad DZ_i = \left(\frac{10}{3}\right)^2$$

$$P(Z_1 + \dots + Z_{150} \leq 480) = P\left(\frac{Z_1 + \dots + Z_{150} - 500}{\sqrt{150} \cdot \sqrt{\frac{100}{3}}} \leq \frac{20}{\sqrt{150} \sqrt{\frac{100}{3}}}\right)$$

$$\stackrel{\text{LFT}}{\approx} \Phi(-0.49) \approx \underline{\underline{31.207\%}}$$

200.3. ~~Това~~ X_i е ~~стандарта~~ δ ~~дискретна~~ и
 Y_i е ~~стандарта~~ δ ~~дискретна~~ по ~~определена~~
 $(X_i, Y_i) = \begin{cases} (1, 0) \\ (0, 1) \\ (-1, 0) \\ (0, -1) \end{cases}$ с ~~вероятност~~ $\frac{1}{4}$.

$$EX_1 = \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot (-1) + 0 \cdot \frac{1}{2} = 0$$

$$EX_1^2 = \frac{1}{4} \cdot 1^2 + \frac{1}{4} \cdot 1^2 = \frac{1}{2} \Rightarrow \boxed{DX_1 = \frac{1}{2}}$$

$$\Rightarrow P(X_1 + X_2 + \dots + X_{1000} > 10)$$

$$= P\left(\frac{X_1 + \dots + X_{1000}}{\sqrt{2} \sqrt{1000}} > \frac{10}{\sqrt{500}}\right)$$

4PT

$$\approx \Phi(0,45) \approx \boxed{32,636 \%}$$

$$d) E[(X_1 + \dots + X_{100})^2 + (Y_1 + \dots + Y_{100})^2]$$

$$= 2E(X_1 + \dots + X_{100})^2 = 2\left(100EX_1^2 + 2\sum_{i \neq j} EX_i X_j\right)$$

$$\begin{aligned} &= 0, i \neq j \\ &X_i \perp X_j \\ &\sim EX_i = 0 \end{aligned}$$

$$= 200EX_1^2 = \boxed{100 = Ed^2}$$

$(X_i, Y_i) \perp (X_j, Y_j)$ for $i \neq j$ u $X_i Y_i = 0$ Beweis!!!

$$E\left((X_1 + \dots + X_{100})^2 + (Y_1 + \dots + Y_{100})^2\right)^2$$

$$= 2E((X_1 + \dots + X_{100})^4) + 2E(X_1 + \dots + X_{100})^2 (Y_1 + \dots + Y_{100})^2$$

$$(X_1^2 + \dots + X_{100}^2 + 2\sum X_i X_j)(Y_1^2 + \dots + Y_{100}^2 + 2\sum Y_i Y_j)$$

$$= 2 \left(100 \mathbb{E} X_1^4 + \binom{100}{2} \binom{4}{2} (\mathbb{E} X_1^2)^2 \right. \\ \left. + 2 \left(100 \mathbb{E} X_1^2 Y_1^2 + 100 \cdot 99 (\mathbb{E} X_1^2)^2 \right) \right)$$

$$= 2 \left(100 \cdot \frac{1}{2} + 50 \cdot 99 \cdot 6 \cdot \left(\frac{1}{2} \right)^2 \right) \\ + 200 \cdot 99 \cdot \left(\frac{1}{2} \right)^2$$

$$\Rightarrow D d^2 = \mathbb{E}(d^2)^2 - (\mathbb{E} d^2)^2 \\ = 9900.$$

* За n стоек: $\mathbb{E} d^2 = n$ и

$$\mathbb{E}(d^2)^2 = 2 \mathbb{E}(X_1 + \dots + X_n)^4 + 2 \mathbb{E}(X_1 + \dots + X_n)^2 (Y_1 + \dots + Y_n)^2 \\ = 2n \mathbb{E} X_1^4 + 2 \cdot \frac{n(n-1)}{2} \cdot 6 (\mathbb{E} X_1^2)^2 + 2 \cdot n(n-1) (\mathbb{E} X_1^2)^2 \\ = n + \frac{3}{2} n(n-1) + \frac{1}{2} n(n-1) = 2n^2 - n$$

$$\Rightarrow D d^2 = 2n^2 - n - n^2 = \underline{\underline{n^2 - n}}.$$

300.4
300.4

$$(a) \mathbb{E}X_n = 2\mathbb{E}X_{n-1} \Rightarrow \mathbb{E}X_n = 2^n$$

$$DX_n = 4DX_{n-1} + 1, DX_0 = 1$$

$$\text{Heva } DX_n =: a_n. a_n = 4a_{n-1} + 1$$

$$a_{n-1} = 4a_{n-2} + 1$$

$$\Rightarrow a_n - 5a_{n-1} + 4a_{n-2} = 0 \quad (\text{Xap. y. e. } t^2 - 5t + 4 = 0)$$

$$t_{1,2} = 4, 1)$$

$$\Rightarrow a_n = C_1 \cdot 4^n + C_2$$

$$a_0 = 1, a_1 = 5$$

$$\Rightarrow \begin{cases} C_1 + C_2 = 1 \\ 4C_1 + C_2 = 5 \end{cases} \Rightarrow C_1 = \frac{4}{3}$$

$$\Rightarrow a_n = \frac{4}{3} \cdot 4^n - \frac{1}{3} = \frac{1}{3}(4^{n+1} - 1) = DX_n$$

$$\delta) \mathbb{E}(X_n - 2X_{n-1})^2 = \mathbb{E}\varepsilon_i^2 = 1$$

$$\Rightarrow \mathbb{E}S_{n,1} = n.$$

$$\mathbb{E}|X_n - 2X_{n-1}|^2 = \mathbb{E}|\varepsilon_i|^2$$

$$\mathbb{E}|N(0,1)| = \int_{-\infty}^{\infty} |x| e^{-\frac{x^2}{2}} dx = 2 \int_0^{\infty} x e^{-\frac{x^2}{2}} dx$$

$$= \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-\frac{x^2}{2}} d\frac{x^2}{2} = \sqrt{\frac{2}{\pi}}$$

$$\Rightarrow ES_{n,2} = n \cdot \sqrt{\frac{2}{\pi}}$$

B) Ako $\varepsilon_i \sim N(1,2)$, to $E\varepsilon_i = 1$ u $D\varepsilon_i = 2$

$$- EX_n = 2EX_{n-1} + 1, EX_0 = 1$$

$$\text{za } a_n = 2a_{n-1} + 1, a_0 = 1, a_1 = 3$$

$$a_{n-1} = 2a_{n-2} + 1 \Rightarrow a_n - 3a_{n-1} + 2a_{n-2} = 0$$

$$\Rightarrow a_n = C_1 \cdot 2^n + C_2 \quad \begin{cases} C_1 + C_2 = 1 \\ 2C_1 + C_2 = 3 \end{cases}$$

$$\Rightarrow EX_n = 2^{n+1} - 1$$

$$DX_n = 4DX_{n-1} + 2 \Rightarrow$$

$$DX_n = C_1 \cdot 4^n + C_2, DX_0 = 1, DX_1 = 6$$

$$DX_n = \frac{1}{3}(5 \cdot 4^n - 2)$$

$$- ES_{n,1} = n E N(1,2)^2 = n \cdot (2^2 - 1) = 3n$$

$$- ES_{n,2} = n E |N(1,2)|^2 = n E [Z\sqrt{2} + 1]^2$$

$$\text{za } Z \sim N(0,1)$$

$$E|2\sqrt{2}+1| = \int_{-\infty}^{\infty} |2\sqrt{2}+1| \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$= \int_{-\frac{\sqrt{2}}{2}}^{\infty} (2\sqrt{2}+1) \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}}_{\phi(x)} dx - \int_{-\infty}^{-\frac{\sqrt{2}}{2}} (2\sqrt{2}+1) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$= \int_{-\frac{\sqrt{2}}{2}}^{\infty} \phi(x) dx - \int_{-\infty}^{-\frac{\sqrt{2}}{2}} \phi(x) dx + \int_{-\frac{\sqrt{2}}{2}}^{\infty} x\sqrt{2} \phi(x) dx$$

$$+ \int_{-\frac{\sqrt{2}}{2}}^{\infty} x\sqrt{2} \phi(x) dx$$

$$= \Phi\left(\frac{\sqrt{2}}{2}\right) - (1 - \Phi\left(\frac{\sqrt{2}}{2}\right)) + \frac{1}{\sqrt{\pi}} \left(\underbrace{\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}}}_{=0} + 2 \int_{\frac{\sqrt{2}}{2}}^{\infty} \right)$$

$$= 2\Phi\left(\frac{\sqrt{2}}{2}\right) - 1 + 2 \int_{\frac{\sqrt{2}}{2}}^{\infty} e^{-\frac{x^2}{2}} d\frac{x^2}{2} \quad \text{or change}$$

$$= 2\Phi\left(\frac{\sqrt{2}}{2}\right) - 1 + \frac{2}{\sqrt{\pi}} e^{-\frac{1}{4}} \approx \underline{\underline{1.3993}}$$

3a $X \sim N(\mu, \sigma^2)$

$$E|X| = \mu \left(2\Phi\left(\frac{\mu}{\sigma}\right) - 1 \right) + \sigma \sqrt{\frac{2}{\pi}} e^{-\frac{\mu^2}{2\sigma^2}}$$

is a folded normal distribution.